

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Cilia (1) Ignore ciliated cell {Move/ sweep/ push/ clear} mucus (1)	2			2		
		(ii)		- An organ is an aggregation of several tissues but this only shows one/ {It/ Image 1} only contains one type of {cell/ tissue} Ignore similar cells	1			1		
	(b)	(i)		Self-replication/ Stage (of division) / stage of development / size/ not fully formed/ age		1		1		
		(ii)		25 x a measurement 1.45/ 1.475/ 1.48/ 1.5 um = 3 marks If incorrect allow (250 x a measurement) for 2 marks 1450/1475/1480/1500 (nm) If incorrect allow a measurement for 1 mark units must be shown 58/59/60mm or 5.8/5.9/6.0 cm		3		3	3	
		(iii)		Explanation must match adaptation Increased/ larger surface area : volume (ratio)(1) Increased gas exchange (1) Cylindrical (1) decreases diffusion distance / large surface area for gas exchange(1) Folding of inner membrane/ cristae (1) {Increased/ large} SA for {enzymes/ membrane proteins / (aerobic) respiration/ / ATP production} (1) NOT energy production/ release	2			2		
				Question 1 total	5	4	0	9	3	0